ABISOLA Z. KEHINDE

Cancer Biology | Metabolic Dependencies | Multi-omics | Precision Oncology

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RESEARCH PROFILE

PhD Candidate in Biochemistry and Molecular Biology and Vanier Canada Graduate Scholar investigating targetable metabolic dependencies in pancreatic cancer. My research combines inducible perturbation systems, mammalian cell models, LC-MS metabolomics, stable-isotope tracing, enzyme assays, molecular biology, and multi-omics to define mechanisms that limit cancer-cell adaptation. I am particularly interested in how tumour-specific dependencies can be translated into biomarkers and therapeutic strategies.

RESEARCH EXPERTISE

Cancer metabolism and tumour dependencies • Mechanistic cell biology • Target validation concepts • Mammalian cell culture and pancreatic cancer organoids • Inducible lentiviral perturbation systems • Cloning, qPCR and western blotting • Flow cytometry • Proliferation and clonogenic assays • Enzyme assays • Drug/nutrient perturbation and rescue studies • LC-MS metabolomics • Stable-isotope tracing • RNA-seq, ATAC-seq and ribosome profiling • R, Python and shell • Experimental design and quantitative data interpretation

RESEARCH EXPERIENCE

Graduate Researcher | Parker Lab, University of British Columbia / BC Children's Hospital Research Institute | Vancouver, Canada | 2021–Present

- Lead a mechanistic cancer-metabolism project asking how acute alanine restriction creates targetable metabolic liabilities in pancreatic cancer cells, integrating perturbation biology with metabolic and growth phenotyping.
- Engineered and validated a doxycycline-inducible alanine-depletion system and used nutrient rescue, inhibitor studies, purified-enzyme assays, and longitudinal cell-growth experiments to separate adaptive pathway changes from growth-limiting effects.
- Mapped adaptive metabolic responses to acute and prolonged nutrient stress with metabolomics and stable-isotope tracing, distinguishing direct pathway rerouting from secondary effects of slowed growth.
- Use mammalian cancer-cell culture, pancreatic cancer organoids, lentiviral systems, molecular cloning, qPCR, western blotting, flow cytometry, proliferation/clonogenic assays, and biochemical assays to test pathway-level hypotheses.
- Develop targeted and untargeted LC-MS and stable-isotope-tracing workflows on an Orbitrap Exploris 240; perform sample preparation, calibration, system-suitability checks, peak review, troubleshooting, and quantitative interpretation.
- Integrate metabolomics with RNA-seq, ATAC-seq, and ribosome-profiling datasets to connect metabolic phenotypes with transcriptional, chromatin, and translational responses.
- Built reproducible R, Python, and shell workflows for normalization, statistical analysis, visualization, and multi-omics integration; developed a time-aware normalization framework for longitudinal metabolomics experiments.
- Present work at national and international meetings, prepare first-author manuscripts, and mentor junior researchers in experimental design, data interpretation, and scientific communication.

Graduate Researcher | PlantSMART Lab, University of British Columbia | Kelowna, Canada | 2019–2021

- Investigated biochemical and compositional phenotypes using microscopy, spectroscopy, enzymatic assays, and quantitative analysis; contributed to experimental design, manuscript preparation, and peer-review responses.
- Led the writing and revision of a first-author study on dietary starch, protein, and fibre composition across breadfruit cultivars and hybrids.

Quality Control Intern | Jawa Pharmaceutical Industry | Lagos, Nigeria | 2015–2016

- Analyzed samples: conducted chemical, physical, and microbiological testing of pharmaceutical products; interpreted test results and assessed their compliance; forwarded the results.
- Verified analytical reports: verified the comprehensiveness of the tests as well as compliance of supporting documents and ensured follow-up of abnormalities.
- Participated in investigations about non-compliances: identified abnormalities and determined the cause; established corrective and preventive actions and ensured follow-up.

EDUCATION

PhD, Biochemistry and Molecular Biology | University of British Columbia, Vancouver, Canada | Expected 2026

MSc, Biochemistry and Molecular Biology | University of British Columbia, Kelowna, Canada | 2021 | Coursework average: 91.8%

BSc, Biochemistry, First Class Honours | University of Lagos, Lagos, Nigeria | 2016 | GPA 4.91/5.00

PhD thesis focus: metabolic vulnerabilities and adaptive nutrient metabolism in pancreatic cancer. Supervisor: Dr Seth J. Parker. MSc thesis: Development of Novel Bioproducts from Breadfruit.

SELECTED PUBLICATIONS AND MANUSCRIPTS

- Kehinde AZ, Parker SJ, et al. Metabolic adaptation to intracellular alanine restriction in pancreatic cancer. First-author manuscript in preparation.
- Kehinde AZ, Parker SJ, et al. Time-aware normalization framework for longitudinal LC-MS metabolomics. First-author manuscript in preparation.
- Johal AS, Al-Shekaili HH, Abedrabbo M, Kehinde AZ, et al. Restricting lysine normalizes toxic catabolites associated with ALDH7A1 deficiency in cells and mice. Cell Reports. 2024;43(12):115069.
- Kehinde AZ, Parker SJ. No Back-up Plan: Loss of Isozyme Diversity as a Promising Therapeutic Strategy for Cancer. Cancer Research. 2022. First author.
- Kehinde AZ, Erland L, et al. South Pacific cultivars of breadfruit and their hybrids have unique dietary starch, protein and fiber. Journal of Food Composition and Analysis. 2022;105:104228. First author.
- Erland L, Needham A, Kehinde AZ, et al. Impact of microclimate on breadfruit fruit and nutritional quality. Journal of Food Composition and Analysis. 2023;115:104983.

SELECTED AWARDS AND HONOURS

- Vanier Canada Graduate Scholarship ($150,000), 2023-2026
- Co-lead $25,000 grant for TOG Indigenous youth mentorship & EDI programming (BC Children's Hospital Research Institute), 2026
- Gordon Research Conference Award ($513), 2025; Gordon Research Seminar Award ($446), 2025; Poster Award, Gordon Research Conference ($161), 2025
- UBC Graduate Student Travel Award ($500), 2025; BCCHR Trainee Travel Grant ($1500), 2024
- UBC Four Year Doctoral Fellowship ($18,200 stipend plus tuition yearly), 2022-2026
- UBC President's Academic Excellence Initiative PhD Award ($1500), 2021-2026
- Best Talk Prize, BMB GSA Summer Symposium ($75), 2024
- UBC Faculty of Medicine Graduate Award ($5931), 2021
- Natural Health Product Research Society of Canada (NHPRS) Student Bursary, 2021
- International Peace Scholarship ($22,000), 2019-2021
- 3MT Final (Three Minute Thesis) People's Choice Award, UBCO ($1000), 2020
- Graduate Dean's Entrance Scholarship ($5000), 2019

SELECTED PRESENTATIONS

- Invited Speaker, Gordon Research Seminar, Feb 2025, Los Angeles, USA
- Poster Presentation, Gordon Research Conference, Feb 2025, Los Angeles, USA
- Invited Talk, CSMB 2024, Manitoba, Canada

- Selected Lightning Talk, BCPM 2024, Vancouver, Canada
- Poster Presentation, UBC Life Science Symposium 2024, Vancouver, Canada
- Virtual Poster Presentation, Cancer Metabolism Showcase 2021
- Talk and Poster, BMB GSA Summer Symposium 2024, Vancouver, Canada
- Selected Talk, CMMT Research Day 2024, Vancouver, Canada
- Invited Speaker, BCCHR OCaD 2023, Vancouver, Canada

LEADERSHIP AND COMMUNITY SERVICE

Chair and Scientific Training Lead, Trainee Omics Group, BC Children's Hospital Research Institute | 2023–2026

- Organized the TOG workshop program, including the inaugural TOG 3MT Data Conference (December 2023), the RNA-seq workshop (November 2024, 100+ trainees), and the omics and HPC workshop on UBC's Sockeye cluster; curated the public workshop recording archive.
- Co-led a successful $25,000 CAD grant supporting Indigenous youth mentorship and equity-focused scientific training.
- Collaborated with three trainee biostatistics/bioinformatics groups to co-host joint events.
- Created and manage a YouTube channel to share recordings from biweekly sessions, extending access to trainees who cannot attend live.

VP Internal, UBC TFFRA | 2024–2025

Vice-President, UBC Muslim Students' Association | 2022–2023; Graduate Student Representative, 2021–2023

Trainee Omics Group Seminar Series Lead, BCCHR | 2021–2023; Trainee Networking Committee Member, 2021–2022

Trainee Representative, BCCHR Scientific Advisory Board | 2024; Student Representative, UBC BMB Head Search Advisory Committee, 2024

TEACHING AND MENTORSHIP

Teaching Assistant, HPC Basics workshop, Precision Health Bootcamp | 2022

- Served as a teaching assistant, supporting trainees with command-line and HPC basics during workshop sessions.

Research Instructor, UBC Vancouver Summer Program | 2024–2026

- Prepared and delivered workshops for international students, guiding the development and presentation of research proposals on clinical trials.

Graduate Teaching Assistant, Biochemistry (BCH 403 Enzymology), UBC | 2022–2023

- Supported lectures, facilitated student discussions, answered student questions, and evaluated assignments and exams.

Graduate Teaching Assistant, Chemistry (CHEM 211 and CHEM 1X3), UBC Okanagan | 2019–2020

- Taught laboratory sessions in person and virtually; graded pre-labs, notebooks, and reports; held office hours. Contributed to online resource development for CHEM 211 (2020).

Mentorship

- Research Mentor, UBC REX Program (mentored 4 undergraduate students; project submitted to CJUR); Future Science Leaders BC (2024–2025); We Guide Learning Initiative (2024–present); Honours Directed Studies students.

WORKSHOPS, OUTREACH AND TRAINING

- Organizer, HPC for Omics Workshop (UBC/BCCHR/ARC; 2025)
- Research Instructor, UBC Vancouver Summer Program (2024–2026)
- Community Outreach, Bella Coola First Nations Science Program (2026)
- Panelist, BCCHR Women in STEM (2024)
- Student Representative, UBC BMB Head Search Advisory Committee, 2024

- Trainee Representative, BCCHR Scientific Advisory Board, 2024
- Volunteer, UBC Centre for Molecular Medicine and Therapeutics External Review, 2024
- Volunteer, BMB Science Rendezvous, 2024
- Volunteer, Discovery Days in Health Sciences, 2023
- Volunteer, Gairdner High School Symposium, 2023
- Volunteer, UBC SMART Program - Let’s Talk Science, 2021-2022
- Volunteer, Phytochemical Society of North America Symposium, 2021
- Volunteer Adjudicator, UBC Multidisciplinary Undergraduate Research Conference, 2021